Device View Tutorial

Vivado provides a tool that allows designs to be viewed after they have been placed and routed in the FPGA. The tool also allows modifications to be made, but this tutorial only focuses on viewing designs. Variations of an adder will be used to demonstrate how the viewer works.

All three files can be loaded into the same Vivado project. Switch the top-level module to implement each part. The three parts are:

1. Implement a combinational 1-bit full adder (fa_1bit.vhd)
2. Add registers to each input and output port of the full adder (fa_1bit_reg.vhd)
3. Implement an 8-bit adder (adder_8bit.vhd)

To view the components, run implementation on the design and then open the implemented design. Vivado should open a window with the design makeup of the FPGA, which has the utilized resources highlighted in white and blue. The routing between all the resources is viewable by making sure the **Routing Resources** button option is selected on the left menu bar. Figure 1 shows an example of the highlighting.

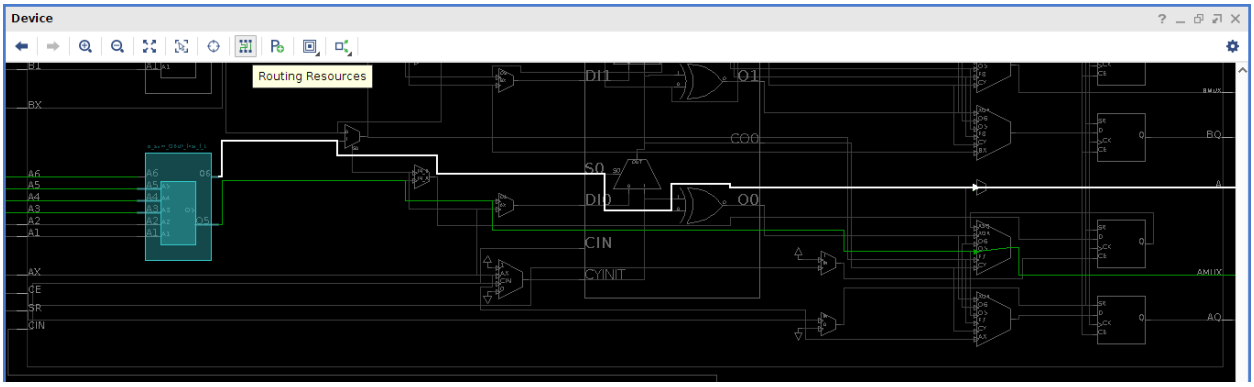


Figure 1: Device View with routing selected

Follow the input signals from the pads to the logic cells, and then follow the outputs signals to the output pads.

For each 1-bit adder implementation, sketch a diagram for each part that represents how it is implemented in the FPGA. For this diagram, only the LUTs and FFs need to be drawn; there is no need to draw buffers, routing multiplexers, etc. However, make sure the equation for each LUT is listed and each flipflop is fully drawn.

The equation and truth table content of each LUT can be viewed by selecting a LUT and navigating to the **Truth Table** tab inside the **Cell Properties** window on the left side. Sometimes two functions can fit inside one LUT6, such as in Figure 1. Click on the outer LUT6 to see one function and click on the internal LUT5 to see the other function.

The FFs within the FPGA have several different configurations, and can be set up as latches or FFs, with active high or low set/reset signals. Depending on the HDL code and Xilinx version, the mapping tools may use the FFs in different ways, so pay close attention to the signals going into each flipflop.

A constrained layout of the resources and their connections can be seen in an easier to read format by viewing the implemented schematic of the design. To do this, select a relevant slice of the design, right-click inside the selection, and select **Schematic**.

Look at the code in the three files. The 1-bit adders were implemented using the Boolean equations for an adder. The 8-bit adder implementation uses the '+' operator. The '+' operator is not defined for a `std_logic_vector` so each input is cast to an `unsigned`. Each input is also concatenated with a leading zero to make room for a possible carry-out bit. The result signal is 9 bits and the lower 8 bits are output on the 8-bit sum port and the most significant bit is output on the carry out port.

For the 8-bit adder implementation, look at the device view and schematic and pay close attention to the CARRY4 blocks, which implement carry chain logic and have dedicated routes to connect portions of the adder across slices.

On the provided SLICEL diagrams, draw the routing necessary for a fully combinational 4-bit adder and the routing for a 4-bit adder with registered outputs. Use *bold red* lines to clearly denote the routes.

This diagram was taken from Figure 2-4 of Xilinx User Guide UG474: "7 Series FPGAs Configurable Logic Block" and more details can be found in that document.

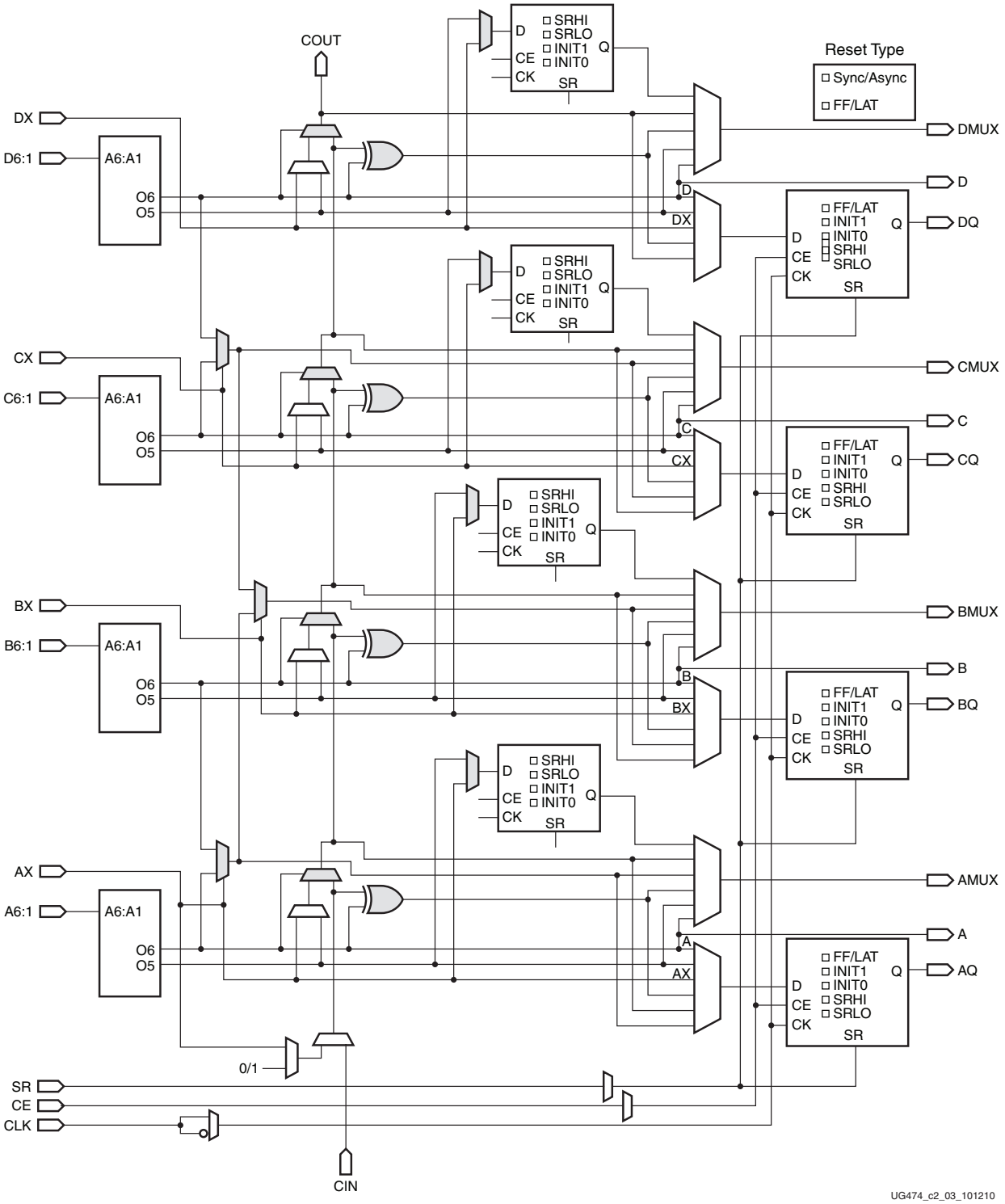


Figure 2: Combinational 4-bit adder routing

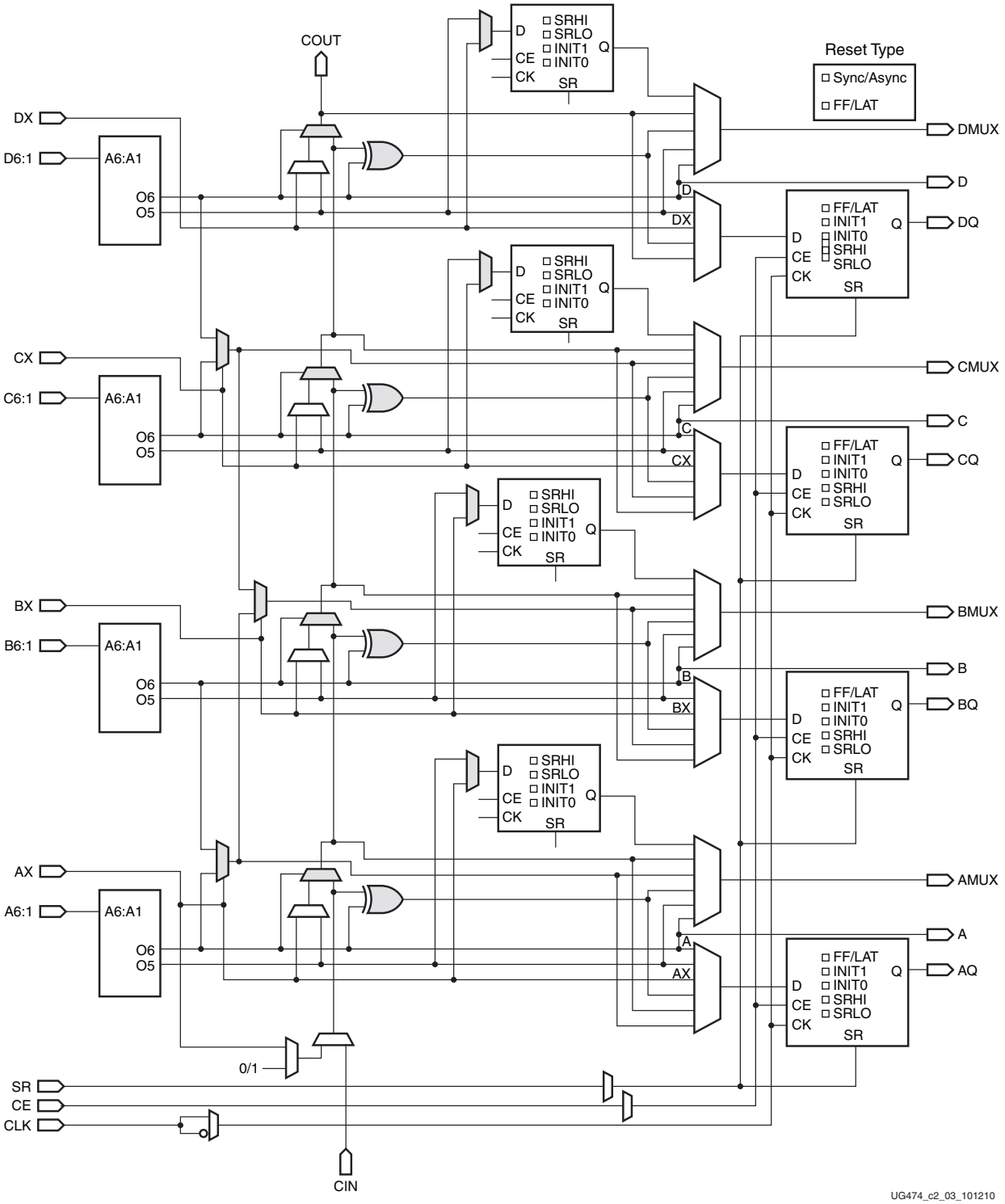


Figure 3: 4-bit adder with registered outputs routing

Tutorial 4: Device View Tutorial

Student's Name:_____

Demo	Point Value	Points Earned	Date
Part 1: Sketched diagram of a combinational 1-bit adder and its implementation in DeviceView	15		
Part 2: Sketched diagram of a registered 1-bit adder and its implementation in DeviceView	15		
Part 3: 8-bit adder in DeviceView	20		
Part 4: 4-bit adder with combinational outputs and its SLICE routing - Figure 2	25		
Part 5: 4-bit adder with registered outputs and its SLICE routing - Figure 3	25		

Students will need to demonstrate their understanding of all parts of this Tutorial by answering questions from Instructor/TAs.

Submit this grading sheet on myCourses under Assignemnts → Tut4.